

[DC] The Proteus Effect in Virtual Reality: Examining Stereotype Threat through Gender Cues

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ABSTRACT

The Proteus Effect and stereotype threat represent two critical phenomena within virtual reality (VR) and social psychology research. The Proteus Effect suggests that embodying virtual avatars can alter self-perception and behavior, while stereotype threat, arising from awareness of negative stereotypes, can impair performance. This project investigates the interplay between these phenomena. Using VR environments, we aim to elicit stereotype threat and assess its impact on cognitive performance and physiological responses. By measuring electrodermal activity (EDA) and heart rate (HR) during working memory tasks, this research seeks to provide objective insights into the psychological and physiological mechanisms of stereotype threat in immersive settings.

Index Terms: Proteus effect, stereotype threat, embodiment, gender identity, virtual reality, social psychology.

1 INTRODUCTION

When people experience stereotype threat, the resulting cognitive dissonance heightens arousal and distress [2]. It leads to increased sympathetic nervous system activation evidenced by elevated cardiovascular response [3] and skin conductance levels [5]. In order for stereotype threat to occur, the negative stereotype must be relevant for the self. This relevance can be established at the individual level or at the group level [11]. Individuals must be aware of the stereotype, but not necessarily believe it, for stereotype threat to occur. On the other hand there is substantial evidence indicating that adopting the identity of another individual results in alterations in self-perception, in direction of our beliefs and stereotypes about that individual, a phenomenon termed the Proteus effect [15, 16]. Furthermore there is evidence that our avatars influence how we interact with others, potentially through behavioral confirmation—a process where one person's expectations lead another to act in ways that confirm those expectations [13].

2 BACKGROUND

The Proteus effect has been observed in various contexts. Yee and Bailenson [15] found that attractive avatars increased openness, while taller avatars boosted confidence in negotiations. Similarly, embodying an obese avatar reduced effort in a sports game [8]. In fully embodied VR, Freeman et al. [4] showed that embodying a short avatar led to negative self-evaluation and paranoia, while Rosenberg et al. [9] found superhero avatars increased prosocial behavior. Seinfeld et al. [10] demonstrated that offenders improved fearful face recognition after embodying a female victim in VR. Originally linked to virtual avatars in games, the Proteus effect extends to cognitive processes, where stereotypes can transfer to the self [14].

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Advancing technology is transforming self-expression, offering immersive opportunities to explore stereotype threat and its impact on social attitudes, especially in VR. In a recent study within the realm of VR, researchers utilized gender-swapped body illusions to counteract the effects of stereotype threat among female participants [7]. Another study investigated whether stereotype threat can be induced among male participants in female virtual bodies. The results supported both the replication and extension hypotheses: female participants in male avatars were protected from stereotype threat, while male participants in female avatars experienced stereotype threat. Furthermore, the gender of the avatar influenced participants' confidence in math and their awareness of the negative stereotype regarding women's math ability [6].

3 RESEARCH GOALS

This project aims to investigate these phenomena by examining how gender swap scenarios in VR influence individuals' experiences of stereotype threat and their subsequent performance on tasks associated with gender stereotypes. Previous studies on stereotype threat in VR have primarily relied on self-reported measures of arousal, which are subjective, can suffer from various types of response biases and do not fully capture the physiological responses involved [1]. Our study will collect objective physiological measures, specifically EDA and HR, as proxies for arousal and cognitive load under stereotype threat. EDA will serve as an indicator of sympathetic nervous system activation, reflecting stress levels, while HR may correlate with task performance. By correlating these measures with performance data (e.g., task accuracy, reaction times), I aim to infer how stereotype threat impacts both emotional regulation and cognitive function. These measures aim to capture the physiological impact of stereotype threat in VR environments, providing a more objective perspective than self-reports alone. I hypothesize that in stereotype threat condition the performance will be impaired as indicated by longer reaction times and more errors while physiological arousal will be higher validated by increased EDA and HR phasic responses.

4 METHODOLOGY

My project will investigate the Proteus effect and stereotype threat using tailored virtual environments for individual and group settings. The research will involve three experiments focused on performance-based tasks using working memory capacity tests.

4.1 First experiment

Research Question1: Can VR environments facilitate elicitation of stereotype threat in women embodying female avatars and in men embodying female avatars?

In the first experiment, participants will engage in working memory tasks within a virtual classroom featuring gender-threatening and non-threatening conditions induced by task framing. Participants in the stereotype threat conditions will see a prompt before starting the tests: "(...)The task on which you are about to participate has been shown to produce gender differences (in no threat condition it will be changed to "not produce gender differences")". I aim to expand upon Peck et al.'s [6] work by using a

different working memory capacity test and incorporating objective measures of arousal. I am planning to use an operation span task (OSPAN), to measure working memory performance. OSPAN involves simultaneously remembering letters while performing math operations [12]. During the whole procedure, I will gather EDA and ECG data to monitor the arousal of participants. Additionally, I will include several questionnaires regarding the embodiment, gender identity and motivation.

4.2 Second experiment

Research Question2: Does exposure to environments dominated by male presence trigger stereotype threat among females, leading to cognitive performance deficits?

The second experiment will try to answer these questions by examining the impact of avatar representation on task performance, specifically focusing on the group setting. I will position participants in a numerical minority status concerning virtual gender, where they will be the sole virtual female or male among two avatars of the opposite gender in a computer-mediated triad. The stereotype threat condition will utilize a design where female participants in female avatars and male participants in female avatars will be outnumbered by male co-actors and engage in a task. In stereotype lift condition male participants in male avatars and female participants in male avatars will be placed with 2 female co-actors. Participants will engage in a mental arithmetic task as proposed by Lee, Nass and Bailenson [?]. This study will incorporate a first-person viewpoint to intensify the body transfer illusion. During the whole procedure I will gather EDA and ECG data and include several questionnaires similar to the first experiment.

4.3 Third experiment

Research Question3: Can gender-specific environmental cues in virtual reality activate stereotype threat and affect behaviors, performance, and self-perception?

In the third experiment, I will investigate whether visual elements typical of a certain gender, such as a room designed to appear masculine or feminine, are sufficient to activate stereotype threat and impact individuals' behaviors, performance, and self-perception when associated with stereotypes. I will incorporate OSPAN to measure if being in an environment typical for opposite gender will impair performance. The hypothesis is that certain situational cues will trigger both objective experiences of identity threat (e.g., increased cognitive and physiological vigilance) and subjective experiences (e.g., reduced sense of belonging and decreased motivation to participate in the environment). These cues may exert significant influence, especially for individuals belonging to stigmatized or stereotype-vulnerable groups, who may be particularly attuned to settings that imply the potential for identity threat.

5 CURRENT STATE

This work is conducted under the supervision of Michał Kuniecki, Emotion and Perception Lab, Institute of Psychology, Jagiellonian University in Kraków. We are utilizing modern equipment to capture both immersive VR experiences and precise psychophysiological data. The setup includes Varjo XR-3 headset and Bitalino measurement device, which enables the recording of ECG and EDA. I am now developing the VR environment and experimental procedure in collaboration with the Faculty of Physics, Astronomy, and Applied Computer Science at Jagiellonian University. This partnership will involve the design and programming of a VR setup tailored to the specific needs of the study, leveraging Unity as our primary development platform.

6 CONCLUSIONS

Questions to mentors:

Q1: How can hand tracking be effectively integrated into a virtual reality setup to enhance participants' sense of embodiment, especially in the absence of haptic feedback? What design considerations can ensure intuitive and immersive interaction with the interface under these constraints?

Q2: What are the key design considerations for creating an effective embodiment procedure in virtual reality experiments? Specifically, how can sensory feedback, avatar customization, and interaction mechanics be optimized to enhance the sense of body ownership and agency?

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